

voice, data, and other types of content to consumers. The base stations are meant to improve the signal frequency and communication between interconnected devices such as computers or smartphones.

Tower: This is where the antennas and electronic communications equipment are placed to create a cell or adjacent cells. The cell towers distribute the signals generated by the base station.

Mobile tower switching office (MTSO): All base stations and towers in a city are connected via a high-speed link or fibre optics to a mobile telephone switching office (MTSO). Our mobile does not have enough signal power to directly contact a caller residing in another city; hence, it sends a signal to a mobile tower. The mobile tower then sends signals to the MTSO. The MTSO checks our SIM data in its database to find the cell in which the phone is present, and sends a signal to the other city's MTSO. Finally, the MTSO sends signals to the mobile through the mobile tower. The MTSO is normally located in the central cell of a cluster and is generally connected to the public switched telephone network (PSTN).

Public switched telephone network (PSTN): A public switched telephone network is a combination of telephone networks used worldwide, including telephone lines, fibre optic cables, switching centres, cellular networks, satellites and cable systems. The PSTN is a century-old worldwide connected telephone network and lets users make landline telephone calls.

Common security concerns: Here is a critical list of security concerns particularly related to the cellular network.

- Traffic monitoring
- Location tracking
- DDOS attacks
- Access fraud
- Stolen phones
- Subscription fraud

Security measures: The key security measures that can help safeguard against

these common security threats are:

- Network traffic monitor and analysis
- Encrypted communication
- Velocity checking
- A subscriber usage-pattern database
- Customer call analysis
- Geographic dispersion checking
- Extensive proprietary anti-fraud algorithms

For more in-depth information about cellular network fraud and solutions, you can refer to <https://csrc.nist.gov/csrc/media/publications/conference-paper/1997/10/10/proceedings-of-the-20th-nissc-1997/documents/031.pdf>.

Wi-Fi

Wi-Fi is the family of wireless network protocols commonly used for local network access. Wi-Fi stands for wireless fidelity. It uses radio frequency to share an internet connection with multiple devices within a certain range via a Wi-Fi router. This router is connected directly to the internet modem and acts as a hub to broadcast the internet signal to all Wi-Fi enabled devices. Wi-Fi repeaters extend the range of the existing network.

Common security concerns: The most relevant security concerns for Wi-Fi networks are:

- Data capture
- On-path attacks
- Denial of service attacks

Security measures: Wired equivalent privacy (WEP) encryption was designed to protect against casual snooping. Later WPS - Wi-Fi protected setup, WPA - Wi-Fi protected access and WAP2 were introduced. All these options are no longer considered secure.

In 2018, WPA3 was announced as a replacement for WPA2, increasing security and enabling secure authentication on wireless routers.

- **WPA3-Personal:** This is a shared key – everyone uses the same key.
- **WPA3-PSK:** The WPA3 session key is derived from PSK using SAE (simultaneous authentication of

equals) to provide stronger defences against password guessing. This allows the access provider and station peers to authenticate each other as part of the handshake process while using cryptographic tools to prevent an attacker from performing an offline password cracking scheme.

Li-Fi

Li-Fi, short for light fidelity, offers an innovative way to connect to the internet using light from lamps, streetlights or LED televisions. Li-Fi uses infrared light or LEDs to transmit data. In addition to being cheaper, safer and faster than Wi-Fi, it does not need a router and just requires you to point your mobile or tablet in direct line-of-sight to the light source to surf the web.

Common security concerns: The major security concerns with Li-Fi technology are:

- Jamming
- Spoofing
- Data modification
- Inability to work without light

Security measures: Li-Fi technology is generally considered to be more secure than Wi-Fi, as light cannot pass through walls like radio waves, and it carries more volume of data at a time. Plenty of security features can be embedded in Li-Fi systems in order to make them more secure, such as encryption and monitoring.

Bluetooth

Bluetooth is a short range communications technology intended to replace the cables connecting portable units while maintaining high levels of security. Each device equipped with a Bluetooth chip communicates via spread-spectrum, frequency-hopping, and a full-duplex signal.

An antenna-equipped chip in each device that wants to communicate sends and receives signals at a specific frequency range defined for short-range communication. To communicate, Bluetooth devices pair with each other